

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Semiconductor Device Theory and Fabrication [ECE_5152]

Marks: 50

Duration: 180 mins.

PART-B

Answer all the questions.

- 1) Explain the Float Zone process of Crystal Growth (5)

A) Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) (5)

A silicon melt has an initial impurity concentration $C_0 = 1.0 \times 10^{20} \text{ atoms/cm}^3$. The equilibrium segregation coefficient for this impurity is $k_0 = 0.35$. Using the Scheil-like result

$$C_s(f) = k_0 C_0 (1 - f)^{k-1},$$

compute the impurity concentration in the solid C_s (in atoms/cm³) at the following fractions solidified:

- (a) $f = 0.10$ (10% solidified)
(b) $f = 0.50$ (50% solidified)
(c) $f = 0.90$ (90% solidified)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Define the segregation constant (distribution coefficient) in crystal growth. Write its expression (2)

- A) and explain the significance of $k < 1$ and $k > 1$ with respect to impurity distribution.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Explain how the thickness of an oxide layer is characterized in semiconductor devices. Describe the commonly used techniques. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (5)

Compare the oxide thickness grown in wet O₂ and dry O₂ at 900 °C after 1 hour for a bare silicon wafer. Given constants:

Wet O₂: $A=0.250\ \mu\text{m}$, $B=0.700\ \mu\text{m}^2/\text{h}$

Dry O₂: $A=0.120\ \mu\text{m}$, $B=0.050\ \mu\text{m}^2/\text{h}$

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Differentiate between Nuclear stopping and electronic stopping (4)

A) Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Write the expression for the total stopping power (S). (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) (4)

A pre-deposition process is carried out for 20 minutes on an n-type silicon wafer having a bulk doping concentration of $1 \times 10^{16}\text{atoms}/\text{cm}^3$ at 1000°C using phosphine (PH₃) as the dopant source. Determine the p–n junction depth x_j assuming a constant (saturated) surface concentration. Given: surface solubility of phosphorus at 1000°C $C_s = 1.0 \times 10^{21}\text{atoms}/\text{cm}^3$, intrinsic diffusivity (D_0) is $0.50\text{cm}^2/\text{s}$, and activation energy $E_a = 3.60\text{eV}$. ($\text{erfc}^{-1}(10^{-5}) \approx 3.123$)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Explain photolithography process step by step with a neat diagram. (4)

A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Describe Contact Printing with its advantages and limitations. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) State the performance Parameters of Lithographic Exposure Tools (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) A 200 nm thick oxide layer is etched using a wet chemical etchant. After 5 minutes of etching, the remaining thickness is 50 nm. (5)

- A) Find:(a) The etch rate in nm/min.(b) The time needed to completely etch the oxide layer.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain the structure and working of a FinFET with the help of a neat diagram. How does it differ from a conventional planar MOSFET in terms of design and performance? (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	12
3	3	CO 3	12
4	4	CO 4	14
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	8	27(54%)
4	Analysis	4	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)